

# Microeconomic Analysis

## DECISION THEORY

Exercises 1-8 correspond to the first tutorial (please submit 1 and 8)  
Exercises 9-16 correspond to the second tutorial (please submit 10 and 14)

### Preferences

1. A problem set of Microeconomic Analysis has  $n$  exercises. One of the students aims for a tutorial work involving less exercises, but two assignments with a difference of exactly one exercise cannot be distinguished. Therefore, the *preferences* of this student can be described by:

$$\text{For every } x, y \in \{0, 1, \dots, n\} : x \succsim y \Leftrightarrow x \leq y + 1$$

- (a) Determine whether these preferences satisfy reflexivity, completeness, connectedness, transitivity, symmetry and antisymmetry.
- (b) Conclude whether these preferences are rational or not.

2. Consider the following questions regarding preferences of a lexicographic nature.

- (a) Consider the Lexicographic Preference on the space of bundles formed by two goods,  $X = \mathbb{R}_+^2$ :

$$(x_1, x_2)L(y_1, y_2) \Leftrightarrow x_1 > y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2)$$

Prove that  $L$  is transitive.

- (b) Consider two individuals, Alice and Bob, that have rational preferences over a finite set of objects  $X$ . A social choice rule is a function that constructs a rational preference for each possible pair of rational preferences  $(\succsim_A, \succsim_B)$  that Alice and Bob may have. Using some reasoning based on the idea of lexicographic preferences, defend (or formally prove), that the following construction is a social choice rule

$$x \succsim y \Leftrightarrow x \succ_A y \text{ or } (x \sim_A y \text{ and } x \succsim_B y)$$

## Utility

3. In Exercise 1, you have probably shown that  $\succsim$  is not transitive by using an example involving three numbers of exercises  $x$ ,  $y$  and  $z$ .<sup>1</sup> Use the same three numbers to show by contradiction that  $\succsim$  cannot be represented by a utility function.

4. Consider the space of bundles  $\mathbb{R}_{++}^2$  and the rational preference  $(x_1, x_2) \succsim (y_1, y_2) \Leftrightarrow x_1^3 x_2 \geq y_1^3 y_2$ .

- (a) What is the simplest utility representation of these preferences that you can think of?
- (b) Discuss whether or not  $v(x_1, x_2) = x_1^{\frac{3}{4}} x_2^{\frac{1}{4}}$ ,  $w(x_1, x_2) = 3 \log x_1 + \log x_2$ ,  $\tau(x_1, x_2) = \frac{1}{x_1^3 x_2}$  are utility representations of these preferences.
- (c) Construct the *uniform consumption equivalent* utility representation in which the utility of  $(x_1, x_2)$  is the amount  $x$  such that the bundle  $(x, x)$  is indifferent to  $(x_1, x_2)$ .

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<sup>1</sup>If not, do it now!

## Revealed Preference

5. A very consistent student in Fordox College has rational preferences  $\succsim$  over the set of meals  $X = \{F(ish), M(eat), P(asta), V(egetable)\}$ . Suppose that your Economics tutor asks you to obtain the preferences by observing the choice behaviour of the student over a week in hall. From Monday to Wednesday, menus were  $\{F, M, P\}$ ,  $\{P, V\}$  and  $\{M, V\}$ .

- (a) Wednesday night, you write an e-mail to your tutor that says: No need to come back tomorrow, job finished! What did the student choose for lunch from Monday to Wednesday?
- (b) Explain which part of the revealed preference has been revealed directly and which indirectly.
- (c) Suppose that, instead, the observed choices were  $P$ ,  $V$  and  $V$ . Since you want to finish your job by Thursday, what menu would you prefer hall to offer next day?

6. Your tutor comes back to you suggesting that Fordox College Hall is a bit more complicated than what was initially expected. The set of meals is indeed  $\{F_H, F_L, M_H, M_L, P_H, P_L, V_H, V_L\}$ , where  $H$  represents high quality and  $L$  represents low quality of the otherwise same meal. Another student has rational preferences  $\succsim$  and you know them to be strictly monotone, in the sense that high quality of the same meal is always preferred to low quality of the same meal. Suppose that, from Monday to Wednesday, you observe menus  $\{F_H, M_H\}$ ,  $\{M_L, P_H\}$  and  $\{F_L, P_L, V_H\}$ , with respective choices  $F_H$ ,  $M_L$  and  $P_L$ . Construct the direct revealed preference and then, the indirect revealed preference (using not only transitivity, but also the knowledge that preferences are monotone). What are the only pieces of information that are missing?

## Rational Choice Data

7. Let  $X$  be finite. Suppose that a universal choice function  $c$  satisfies the Weak Axiom of Revealed Preference. That is, you cannot find two different alternatives  $x, y$  such that  $x \succ_c y$  and  $y \succ_c x$  is revealed in two different menus. Prove formally that  $c$  is rational.

8. In a letter to the New York Times, a shoe salesman wrote:

*as a neophyte shoe salesman, I was told never to show customers more than three pairs of shoes. If they saw more, they would not be able to decide on any of them.*

Consider a shop with a stock of shoes  $A, B, C$  and  $D$  and denote by  $\diamond$  the no-pair-sold situation in which the client leaves the shop without buying any pair of shoes.

- (a) Discuss the form of the choice function of an almost rational client that always chooses according to a preference  $\succsim$  over  $A, B, C, D$ , except when the four pairs of shoes are offered, as suggested by the quote.
- (b) Demonstrate that the choice function fails to satisfy property  $\alpha$ , and the weak axiom of revealed preference. Is this behaviour rational?
- (c) Suppose that the shop makes 5 pounds with any shoe sold. Moreover, the shop makes extra 15 pounds (say, in future sales), if the shoe sold is the best according to the preference  $\succsim$  of the client, extra 10 pounds if the shoe is the second best, and extra 5 pounds if it is the third best. The shop makes 0 pounds if no shoe is sold. Suppose that a random client arrives. Argue that, when the client arrives, it is in the interest of the seller, at least weakly, to show exactly three pairs of shoes.
- (d) If half of the clients have the preference  $A \succ B \succ C \succ D$ , 30 per cent have preferences  $B \succ C \succ D \succ A$  and 20 per cent have preferences  $D \succ A \succ B \succ C$ , which three pairs of shoes would you show and what would be the expected profit per client?

## Economic Frameworks

9. Consider the space of bundles  $\mathbb{R}_+^2$ . A consumer has strictly monotone preferences and spends all income in the two goods considered. We observe the choices made in two situations, when prices for goods are  $(p_1, p_2) = (3, 1)$  and when prices for goods are  $(1, 3)$ .

- (a) Show that observed choices cannot be  $(x_1, x_2) = (2, 1)$  in the first case, and  $(1, 2)$  in the second. What if the choices are  $(2, 1)$  and  $(\frac{7}{4}, \frac{7}{4})$ ?
- (b) Show that observed choices can be  $(1, 3)$  in the first case and  $(1, 2)$  in the second.
- (c) In the latter case, discuss whether preferences can be strictly convex or not, and whether they can be Cobb-Douglas or not

10. Consider a rational individual that has the following preferences over a set of timed consumptions (the first component capturing money, and the second the time at which it is paid)

$$(9, 0) \succ (9, 1) \sim (4, 0) \succ (9, 2) \succ (4, 1) \succ (4, 2) \sim (1, 0) \succ (1, 1) \succ (1, 2)$$

- (a) Show that, although the individual is rational, there is no discounted utility that explains these preferences (and hence, that would explain the choices generated by them). What property is not satisfied?
- (b) Fix the utility function over money  $u(x) = \sqrt{x}$ . Show that you can find a sort of impatience parameter  $\gamma \geq 0$  such that the utility function  $U(x, t) = \frac{1}{1+\gamma t} u(x)$  can represent these preferences. Discuss this representation.

## Probability

11. Discuss, using Jensen's inequality, whether the following inequalities are true or not

(a)  $\prod_{i=1}^n x_i^{\lambda_i} \leq \sum_{i=1}^n \lambda_i x_i$

(b)  $(\sum_{i=1}^n x_i)^2 \leq n \sum_{i=1}^n x_i^2$

12. A population of voters have single-peaked preferences over the set  $[-1, 1]$ . The density function  $f(z) = cz^2$  describes the mass of individuals with peak at policy  $x \in [-1, 1]$ .

- (a) What is the value of the constant  $c$ ?
- (b) Write down the formulas for the expected value and the variance of the distribution of peaks. Compute them using these formulas. What is the median of the distribution (and hence, the optimal policy with majority voting)?
- (c) What fraction of voters have a peak larger than  $\frac{1}{2}$ . Use this example, maybe in conjunction with the uniform distribution, to informally discuss how to formalise the idea of polarisation of political opinions within a population.

### Expected Utility

13. Let  $Z$  be a finite set of prizes.

- (a) Define formally one rational preference over lotteries satisfying continuity but not independence.
- (b) Define formally one rational preference over lotteries satisfying independence but not continuity.

14. Consider lotteries, all of them with payoffs in  $[0, 1]$ , that have density functions of the form  $f(z) = (\beta + 1)z^\beta$ , with  $\beta \in [0, 1]$  (notice that  $\beta = 0$  corresponds to the uniform distribution).

- (a) Show that the expected value of the lotteries is increasing in  $\beta$ . Can you easily conclude from there that the best lottery is that with  $\beta = 1$ ? Can you maybe use an argument based on first order stochastic dominance?
- (b) An individual has risk averse CRRA preferences  $z^{1-r}$ , with a risk aversion parameter  $r \in [0, 1]$ . Compute the Expected Utility and the Certainty Equivalent of each lottery.

15. (Adapted from Casino, Osborne and Rubinstein). An individual has expected utility preferences. The individual has wealth  $w$  and has to choose an amount  $x \leq w$ , after which a lottery is conducted in which with probability  $\alpha$  she gets  $2x$  and with probability  $(1 - \alpha)$ , she loses  $x$ .

- (a) Show that, if the individual is risk averse, the higher the value of  $\alpha$ , the higher the amount that she chooses.
- (b) Suppose that the individual has CRRA preferences. Compute the amount chosen and show that it decreases with the level of risk aversion of the individual.

16. Suppose that two individuals have the same initial wealth and CRRA utility functions  $u_1 = z^{1-r_1}$  and  $u_2 = z^{1-r_2}$ , with  $r_1 < r_2$ .

- (a) Compare the Arrow-Pratt coefficients of risk aversion of both individuals
- (b) From the Arrow-Pratt theorem, you can then deduce that the utility function of individual 2 is a concave transformation of that of individual 1. Prove it directly to confirm.
- (c) Use also the Arrow-Pratt theorem to explain why, if both face the same lottery  $p$  and individual 1 rejects it, individual 2 also reject it